For the system $\mathrm{d}z/\mathrm{d}t = \boldsymbol{f}(z)$, a coordinate transformation $z \mapsto \tilde{z} = \boldsymbol{R}z$ results in

$$\frac{\mathrm{d}\tilde{z}}{\mathrm{d}t} = \boldsymbol{R}\frac{\mathrm{d}z}{\mathrm{d}t} = \boldsymbol{R}\boldsymbol{f}(\boldsymbol{R}^{-1}\tilde{z}) = \boldsymbol{R}\boldsymbol{f}(\boldsymbol{R}\tilde{z}),$$

since $\boldsymbol{R}^{-1} = \boldsymbol{R}$.

A change of time $t \mapsto \tau = -t$ results in

$$\frac{\mathrm{d}z}{\mathrm{d}\tau} = \frac{\mathrm{d}t}{\mathrm{d}\tau}\frac{\mathrm{d}z}{\mathrm{d}t} = -\boldsymbol{f}(z).$$

In other words, for a reversible vector field, the coordinate transformation $z \mapsto \boldsymbol{R}z$ is equivalent to the change of time $t \mapsto -t$.

3.6.2 Time-Reversal Symmetry and Symmetric Numerical Methods

The flow map $\mathscr{F}_t(z_0)$ gives a solution of the differential equation, thus for a time-reversible system, we have, with $z(t) = \mathscr{F}_t(z_0)$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{F}_t(z_0) = \frac{\mathrm{d}z}{\mathrm{d}t} = -\boldsymbol{R}\boldsymbol{f}(\boldsymbol{R}\mathscr{F}_t(z_0)).$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{R}\mathscr{F}_t(z_0) = -\boldsymbol{f}(\boldsymbol{R}\mathscr{F}_t(z_0)),$$

and if we define $\tilde{z}(t) = \boldsymbol{R}\mathscr{F}_t(z_0)$ then $\tilde{z}(t)$ satisfies the differential equation

$$\frac{\mathrm{d}\tilde{z}}{\mathrm{d}t} = -\boldsymbol{f}(\tilde{z}).$$

From this we see that if we start from a point z_0 and integrate forward in time (i.e apply $\mathscr{F}_t$) then apply the involution $\boldsymbol{R}$, we get the exact same result as if we start from $\tilde{z}(0) = \boldsymbol{R}z_0$ and integrate *backward* in time (i.e apply $\mathscr{F}_{-t}$). In sum, since z_0 is an arbitrary point,

$$\mathscr{F}_{-t}(\boldsymbol{R}z) = \boldsymbol{R}\mathscr{F}_t(z).$$

For the flow map, $\mathscr{F}_{-t} = \mathscr{F}_t^{-1}$ so we can recast the identity as

$$\boldsymbol{R} \circ \mathscr{F}_t \circ \boldsymbol{R} \circ \mathscr{F}_t = \mathrm{Id}, \tag{3.14}$$